

Abdullah Ayman

+20 114 351 2531 | abdullahayman@gmail.com | abdullahayman.netlify.app | github.com/BodeAyman | linkedin.com/in/abdullah-ayman9

EDUCATION

Zewail City of Science and Technology

Bachelor of Science in Computer Science | GPA: 3.92/4.0 *Expected July 2027*

- **Relevant Coursework:** Operating Systems, Software Architecture, Databases, Data Structures, Algorithms, OOP, Web Development, Machine Learning, Object-Oriented Analysis and Design

EXPERIENCE

Founding Engineer (Full-Stack)

Sep. 2025 – Present

Moka | Play Store

Remote

- Led full-stack development of **Ma3ak**, a construction labor marketplace (Flutter, React, .NET Core), reaching **50 active users** within 6 weeks using PostgreSQL and MongoDB for hybrid data modeling
- Implemented mobile-side caching, Redis-backed backend services, WebSocket-based real-time chat, and Firebase Cloud Messaging (FCM) notifications for improved responsiveness and user engagement
- Containerized backend services with Docker behind Nginx reverse proxy; built a GitHub Actions CI/CD pipeline reducing deployment time from **3 hours to under 2 minutes**, enabling zero-downtime releases

Open Source Contributor

Mar. 2026 – Present

istSOS4 · OSGeo | GitHub

Remote

- Merged 3 security-critical patches into production, eliminating SQL injection and schema leakage vulnerabilities across **14+ affected API endpoints**
- Submitted 2 backend PRs resolving ID sequencing failures, invalid query edge cases, and non-standardized error responses to harden API reliability
- Shipped 2 Next.js features: resolved entity editing workflow bug and built a Time-Travel History & Commit Explorer enabling historical sensor data queries at any timestamp

PROJECTS

ClipSphere | Next.js, Node.js, MongoDB, MinIO | GitHub

- Engineered a full-stack social media platform featuring JWT authentication, role-based access, post feeds, likes, comments, and user-to-user interactions
- Implemented MinIO-based object storage for scalable media handling and optimized backend APIs with Node.js and MongoDB for low-latency content delivery
- Developed responsive UI with Next.js, integrated WebSocket-based real-time messaging/notifications, and established the backend architecture foundation

FloosFlow | Express.js, React, PostgreSQL | GitHub

- Built P2P payment platform with ACID-compliant PostgreSQL transactions ensuring data integrity across concurrent transfers; WebSocket subscriptions eliminated polling, cutting notification latency by **~60%**
- Integrated Google OAuth 2.0 for frictionless authentication and persistent WebSocket subscriptions delivering real-time balance updates with sub-**100ms** latency

Lazeez | Express.js, MongoDB, RabbitMQ | GitHub

- Built 3-microservice recipe discovery API (Express.js, MongoDB) communicating asynchronously via RabbitMQ, eliminating synchronous bottlenecks and achieving sub-**200ms** average response time under concurrent load
- Implemented multi-dimensional budget and calorie filtering with compound MongoDB indexing, reducing query time by **~40%** versus unindexed baseline

ACHIEVEMENTS & COMPETITIONS

Egyptian Collegiate Programming Contest (ECPC)

Certificate | Aug. 2025

Ranked 21st in Qualifiers — Advanced to ECPC Finals

Egyptian Collegiate Programming Contest (ECPC)

Certificate | Jul. 2024

Honorable Mention

Central Bank of Egypt FinYology Hackathon

Certificate | May 2025

Finalist — nutrition-tracking fintech app (Node.js, React) linking meal costs to real-time budget limits

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C++, C#, Dart, PHP, Java

Frameworks & Libraries: React, Next.js, Express.js, Node.js, .NET Core, Flask, FastAPI, Flutter, Prisma

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis

Developer Tools: Git, Docker, GitHub Actions, Postman, Swagger

Infrastructure & Protocols: Nginx, RabbitMQ, WebSockets, GraphQL